Distillation De-Virus: Mathematical Roots and Multi-Scenario Generalization

Date: 2026-06-26 | SCX v4.0 | From “Data Poisoning Defense” to “Distillation De-Virus”: the mathematical foundations

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1. Problem Definition: Why Is Distillation “Poisoned”?

1.1 Standard Knowledge Distillation Review

The framework of Hinton et al. (2015):

Teacher $f_T : X \to Y$, trained on training set $D_{\text{train}} = \{(x_i, y_i^*)\}$.

Distillation: Teacher generates soft or hard labels for a set of unlabeled data $\{x_i^{\text{distill}}\}$:

```
y_i^distill = f_T(x_i^distill)
```

Student $f_S : X \to Y$, trained on $D_{\text{distill}} = \{(x_i^{\text{distill}}, y_i^{\text{distill}})\}$.

1.2 The Mathematical Essence of the Problem

The core problem is: **the Teacher is not perfect**. Define the Teacher’s state-conditioned risk:

```
R_T(s) = E_{x~P(·|s)} [ (f_T(x), f*(x)) ]
```

where $f^*(x)$ is the true label (unknown), and s is the state.

Core observation: $R_T(s)$ varies greatly with s. In AlN v3 training:

State s (configuration region)	$R_T(s)$ (force RMSE, eV/Å)
Phonon region	0.008
EOS near-equilibrium region	0.015
Elastic deformation region	0.011
Thermal 1800K region	**0.077**
MLMD stress10 region	**0.036**

The Teacher's error in the thermal region is **9.6**× that of the phonon region.

Distillation Poisoning Mechanism:

```
Teacher generates distillation data in thermal 1800K region
    → Distillation labels y_distill = f_T(x) = f*(x) + _T(s)  [_T is teacher bias
    → Student fits: f_S(x)  y_distill = f*(x) + _T(s)
    → Student error in thermal region  _T(s)
    → Student "inherits" systematic bias from Teacher
```

The difference from VASP data poisoning:

```
VASP Poisoning: f*(x) itself is wrong (DFT labels have problems)
Distillation Poisoning: f*(x) is correct (DFT labels are right), but Teacher did r
```

Commonality: Both are state-conditioned—poisoning only occurs in specific configuration regions.

1.3 Why Distillation Poisoning Is Harder to Diagnose Than VASP Poisoning

	VASP Data Poisoning	Distillation Data Poisoning
Diagnosis signal	$f_{\max} > 5$ eV/Å (directly observable)	No direct signal
Label source	DFT (known reliability)	Teacher (unknown reliability)
Interpretability	Yes—wrong configuration → wrong forces	**No**—distillation outputs look "normal"
Prior knowledge needed	Know that physically $f_{\max}$ should be < 1 eV/Å	Need to know how reliable Teacher is in each region
SCX's role	Automatically detect anomalous $f_{\max}$	Use Teacher's residual map on training set to infer

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2. Mathematical Analysis of Existing Methods

2.1 Hinton KD (2015)—No Purification, Full Distillation

Approach: $T \to S$, all distillation samples treated equally.

Implicit assumption: Teacher is equally reliable across all input regions.

Mathematically: This is equivalent to assuming $\mathrm{Var}[\ell(f_T(x), f^*(x))] \approx 0$, i.e., the variance of teacher error is negligible. But in MLIP scenarios, Teacher error across different configuration regions can differ by an order of magnitude—this assumption clearly does not hold.

2.2 Confidence-Based Filtering

Approach: Use Teacher's output entropy $H(f_T(x))$ or maximum softmax value as "confidence," filtering low-confidence distillation samples.

Mathematical form:

```
D_clean = {x_i | max_j f_T^{(j)}(x_i) > _confidence}
```

Why it fails in MLIP scenarios:

Teacher (ACE linear model) outputs energy E and forces F, not a probability distribution. There is no "confidence" concept—E and F are deterministic outputs. **Confidence filtering is directly inapplicable.**

Even in classification scenarios: high confidence $\neq$ low error. A model can be highly confident about a wrong prediction (overconfidence). Confidence does not reflect state-conditioned reliability structure.

2.3 Ensemble Disagreement Filtering

Approach: Train M Teachers (different initializations/data subsets), use output variance as uncertainty:

```
²_ens(x) = Var[f_T1(x), f_T2(x), ..., f_TM(x)]
D_clean = {x_i | ²_ens(x_i) < _ens}
```

Advantages: More reliable than confidence; captures model uncertainty across regions.

Limitations: Requires training M Teachers $\rightarrow$ high cost (M ACE trainings $= M \times 40$ epochs in MLIP). Ensemble variance reflects "consistency among multiple Teachers," not "error between Teacher and true labels." If all Teachers make the same systematic error in the same region (common—due to architectural limitations), ensemble variance will be low but actual error large.

2.4 Data Shapley / LOO Values

Approach: Compute Shapley values or LOO error for each training sample; retain high-value samples, discard low-value ones.

Mathematically: Shapley value for the i-th sample computes "the marginal contribution of this sample to model performance." But this is a **global quantity**—it does not distinguish contributions across different states.

Limitations: Like confidence, Shapley values contain no state-conditioned information. A sample contributing to the phonon region and one contributing to the thermal region may have identical Shapley values, but Teacher reliability in the thermal region is completely different.

2.5 Fundamental Deficiency of All Methods

Method	Signal needed	MLIP applicable?	Distinguishes Teacher regional reliability?	Distinguishes state conditions?
Confidence filtering	softmax/logits	(ACE has no probabilistic output)		
Ensemble disagreement	M-model variance	(feasible but costly)	Partial	
Data Shapley	All-subset retraining	(computationally infeasible)		
SCX De-Virus	**Teacher's residual on $D_{\mathbf{train}}$ + Encoder**			

All existing methods fail to distinguish states—they treat Teacher reliability as a homogeneous problem solvable at the sample level (rather than state level). SCX's core contribution is pointing out that this assumption is wrong.

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3. SCX Distillation De-Virus Mathematical Framework

3.1 Core Idea

Not judging distillation data quality sample by sample, but judging Teacher reliability at the state level.

Intuition: Sample-level judgment: Is this distillation frame's force label correct? $\rightarrow$ Unanswerable, because there is no ground truth. State-level judgment: What is the overall reliability of the Teacher in the "thermal 1800K configuration region"? $\rightarrow$ Answerable, because Teacher already shows high error on thermal test frames from the original training set.

3.2 Formal Definition

Known (from Teacher training): $D_{\text{train}} = \{(x_i, y_i^*)\}$ — original training set with DFT labels. Encoder $\varphi : \mathcal{X} \to \mathbb{R}^d$ — State Encoder. Teacher f_T already trained.

Computable (no additional DFT needed): Teacher's residual on each sample in D_{train}: $r_i = \ell(f_T(x_i), y_i^*)$. State discovery: cluster with $\varphi \rightarrow$ state partition $\mathcal{S} = \{s_1, \ldots, s_K\}$. Teacher reliability per state:

```
R_T(s_k) = (1/|s_k|) _{x_is_k} (f_T(x_i), y_i^*)
```

This is the **Teacher Reliability Map**—essentially a map of "which configuration regions the Teacher is unreliable in."

Distillation data assessment: $D_{\text{distill}} = \{(x_j^{\text{distill}}, f_T(x_j^{\text{distill}}))\}$. For each x_j^{distill}, encode $\varphi(x_j^{\text{distill}})$, assign to nearest state s_k. If $R_T(s_k) > \tau$, mark x_j^{distill} as "virus frame."

Filtered distillation:

```
D_clean = {x_j^distill | R_T(state(x_j^distill))  }
```

Train Student f_S on D_{clean}.

3.3 Mathematical Guarantee (Heuristic)

Claim: Let Teacher bias in state s be $\delta_T(s) = |\mathbb{E}[f_T(x) - f^*(x) \mid x \in s]|$. Student trained on filtered distillation set satisfies, for expected bias in state s:

```
E[|f_S(x) - f*(x)| | xs]  _S_clean(s) +  · max(0, _T(s) - )
```

where $\delta_{S,\text{clean}}(s)$ is Student's intrinsic learning error on clean data, and the second term is the leakage of teacher bias.

Interpretation: By setting τ to filter high-bias states, we can control the propagation of Teacher's systematic error to the Student. Stricter filtering (smaller τ) means less leakage but fewer distillation samples.

Trade-off:

```
V_s() = |D_clean()| / |D_distill|  - retention ratio
Q_s() = 1 - (_T(s) - )_+ / _T(s)  - quality improvement ratio
```

The optimal τ depends on: distribution of Teacher reliability across states, coverage requirements of distillation data, and Student sensitivity to data volume.

3.4 Why Two-Layer Descriptors Are Especially Critical Here

Single-layer Encoder (e.g., MLIPEncoder 12-dim): Coarse granularity, 50% of frames squeezed into one state. Cannot distinguish "EOS near-equilibrium frames" from "thermal 1800K frames" (they may be grouped into the same coarse state). → One state mixes Teacher-reliable and Teacher-unreliable sub-regions. → Filtering either "wrongly kills" reliable frames or "lets through" unreliable frames.

Two-layer Encoder (Layer 1 + ErrorDrivenEncoder): Layer 1 coarse clustering: large-region partitioning. Layer 2 error-driven fine clustering: within each Layer 1 cluster, re-cluster using feature dimensions correlated with Teacher error. → Separates Teacher-unreliable sub-regions from reliable ones. → Precise filtering: only discard sub-states where Teacher is genuinely unreliable.

This is the unique value of the two-layer method in distillation de-virus—it can separate "Teacher's reliable regions" and "Teacher's unreliable regions" within the same coarse region.

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4. Distillation De-Virus Across Different Scenarios

4.1 Scenario 1: MLIP Potential Distillation (ACE → ACE/NEP)

Distillation approach: Teacher ACE potential computes single-point energy + forces on many configurations → generates (configuration, energy, force) data → trains Student NEP / lightweight ACE.

Poisoning risk: Teacher's high error in extreme regions (thermal 1800K / MLMD stress10) propagates to Student.

SCX Solution: 1. Teacher trained on D_{train} (534 frames). 2. Compute Teacher per-state force RMSE from D_{train} → Reliability Map. 3. Distillation generates 5000 frames (Teacher performs single-point calculations on ASE-rattled configurations). 4. SCX Encoder encodes distillation frames → maps to states. 5. Filter distillation frames from thermal 1800K and MLMD stress10 regions. 6. Train Student on clean distillation set.

Encoder choice: `MLIPEncoder` (12-dim handcrafted descriptors), refined with two-layer method.

4.2 Scenario 2: Image Model Distillation (ResNet-50 → MobileNet)

Distillation approach: Teacher ResNet-50 infers on unlabeled images → (image, soft label) → trains MobileNet.

Poisoning risk: Teacher recognition errors in specific image subspaces (e.g., low light, occlusion, blur) propagate to Student.

SCX Solution: 1. Teacher trained on CIFAR/ImageNet training set. 2. Compute Teacher per-state classification error from validation set → Reliability Map. 3. Distill many unlabeled images. 4. `VisionEncoder` (CNN backbone) encodes → maps to states. 5. Filter distillation frames from Teacher high-error states (e.g., blurry image regions). 6. Train Student on clean distillation set.

Encoder choice: `VisionEncoder` (simple_cnn/resnet backbone).

Key difference from confidence filtering: Confidence filtering only looks at Teacher's softmax output ("how confident is Teacher itself?"). SCX looks at Teacher's actual prediction error on similar images ("does Teacher really err a lot in this region?").

4.3 Scenario 3: LLM Distillation (GPT → Small Model)

Distillation approach: Teacher LLM generates many QA pairs → (prompt, response) → trains Student small model.

Poisoning risk: Teacher hallucinations/errors in specific semantic subspaces (e.g., reasoning problems, code generation) are amplified and propagated through distillation data.

SCX Solution: 1. Teacher evaluated on benchmark datasets. 2. Use evaluation results to build Teacher per-topic reliability → Reliability Map. 3. Distillation generates many QA pairs. 4. `LLMEncoder` (BERT/GPT embedding) encodes prompts → maps to semantic states. 5. Filter distillation QA pairs from Teacher high-error semantic states (e.g., multi-step reasoning). 6. Train Student on clean distillation set.

Encoder choice: Needs implementation of new `LLMEncoder`—use sentence embeddings (e.g., BERT) to encode prompts as vectors, cluster in semantic space.

Unique value: This is a scenario where even confidence filtering struggles—LLMs are often highly confident about wrong answers (hallucination with high confidence). SCX does not rely on Teacher's self-assessment; it characterizes reliability using actual performance on external benchmarks.

4.4 Scenario 4: Simulation Fidelity Distillation (Fine-mesh FEM → Coarse-mesh + Surrogate)

Distillation approach: Fine-mesh FEM samples parameter space → (parameter, response) → trains reduced-order model / coarse-mesh correction term.

Poisoning risk: When fine-mesh FEM's numerical accuracy degrades in specific parameter regions (e.g., near resonance, near bifurcation), distillation data carries numerical error.

SCX Solution: 1. Fine FEM evaluated on verification points. 2. Use verification error to build FEM per-region reliability → Reliability Map. 3. FEM densely samples parameter space. 4. `TabularEncoder` (simulation parameter vector) encodes → maps to parameter states. 5. Filter simulation points from FEM high-error parameter regions (e.g., near-bifurcation regions). 6. Train Student reduced-order model on clean simulation set.

Encoder choice: `TabularEncoder` (simulation parameter vector, e.g., Re, Ma, load amplitude...).

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5. Relationship to Existing SCX Work

```
Unified view of SCX's four capabilities:


           SCX = State-Conditioned Data Value Judgment Framework


   Data Poisoning Defense
     Judge: fmax (directly observable from VASP)
     Output: Flag bad VASP frames
     AlN v3 Verification: F1=0.585, 100% hit rate

   Redundancy Compression
     Judge: Current model error in state s
     Output: Low-error high-density regions compressible
     AlN v3: phonon region 50-80% compressible

   Distillation De-Virus ← New
     Judge: Teacher reliability in state s (from D_train)
     Output: Filter distillation data from Teacher-unreliable states
     Principle: Generalization of -from VASP bad frames
               to Teacher bad predictions

   Expert Routing
```

```
Judge: Reliability differences among multiple experts
       in state s
Output: Route each state to the most reliable expert
Principle: Generalization of -not just filtering,
          but also selecting the best
```

Unified mathematical core: All four capabilities are built on estimating "state-conditioned reliability." The only difference is the **source of the judgment signal**:

Capability	Judgment Signal	Signal Source
Data Poisoning Defense	$f_{\max}$ (VASP output)	DFT calculation
Redundancy Compression	Current model error	Training set
Distillation De-Virus	Teacher error on D_{train}	Teacher training history
Expert Routing	Multi-expert reliability differences	Cross-expert comparison

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6. Literature Survey: Quality Control in Knowledge Distillation

Existing Research Directions

Direction	Representative Work	Difference from SCX
Confidence-based	Balan et al. (2022), "Confidence-Aware KD"	Confidence $\neq$ accuracy; inapplicable to deterministic models (e.g., ACE)
Ensemble-based	Malinin et al. (2020), "Ensemble Distillation"	Requires M Teachers; ensemble variance $\neq$ true error
Data valuation	Ghorbani & Zou (2019), "Data Shapley"	Global quantity; does not distinguish state conditions
Curriculum KD	Jin et al. (2019), "Curriculum KD"	Sorted by difficulty, not by Teacher reliability

Direction	Representative Work	Difference from SCX
Contrastive KD	Tian et al. (2020), "CRD"	Contrastive learning improves representation, does not address local Teacher unreliability
Online KD	Zhang et al. (2018), "Deep Mutual Learning"	Mutual learning, does not distinguish which region Teacher is strong in
SCX De-Virus	**This paper**	**State-conditioned Teacher reliability → filter unreliable regions → purify distillation data**

Core Gap

The common assumption of all existing KD quality control methods: **Teacher reliability can be judged at the sample level (per-sample)**. SCX's unique contribution is pointing out that this assumption is false—reliability is state-level (per-state), because Teacher error is systematically concentrated in specific configuration/feature regions.

SCX is the first work to introduce the concept of "state-conditioned expert reliability" into knowledge distillation.

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7. Implementation Roadmap

Phase 1: MLIP Verification (AlN v3, 2–3 days)

```
1. Train Teacher (Single ACE) on AlN v3 534 frames
2. Compute Teacher per-state force RMSE → Reliability Map
3. Teacher distills 5000 frames (single-point energy+forces on ASE rattled configu
4. SCX two-layer analysis of distillation data
5. Filter Teacher high-error states (thermal 1800K, MLMD stress10)
6. Train Student A (full distillation set) vs Student B (de-virus distillation set
7. Compare force RMSE of both on thermal/MLMD test sets
```

Expectation: Student B has significantly lower force RMSE in the thermal region than Student A.

Phase 2: Image Verification (CIFAR-100, 1 week)

```
1. Teacher ResNet-50 → MobileNet-v2 distillation
2. SCX uses validation set to establish Teacher per-class/per-feature-region relia
3. Filter distillation data from Teacher high-error regions
4. Compare: Full KD vs Confidence-filtered KD vs SCX-filtered KD
```

Phase 3: LLM Verification (Needs LLMEncoder Implementation)

Phase 4: Public Dataset Cleaning (Community Contribution)

```
MPTraj / OC20 / ColabFit → SCX two-layer analysis → SCX-cleaned → GitHub Release
```

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8. Summary of Core Insights

1. **Distillation poisoning is state-conditioned**—Teacher error is not uniformly distributed; it concentrates in specific configuration/feature regions. This is the fundamental reason SCX can solve the problem.
2. **All existing methods judge at the sample level**—confidence, ensemble variance, Shapley values are all per-sample. They do not leverage the information that "Teacher is overall unreliable in state s."
3. **SCX's unique value is elevating KD quality control from sample-level to state-level**—using Teacher's per-state reliability map from the training set instead of per-sample confidence for structured data filtering.
4. **Two-layer descriptors are the core of precise filtering**—single-layer coarse clustering mixes Teacher-reliable and unreliable sub-regions; two-layer refinement separates them in the error subspace.
5. **This is a paper-level contribution**—"State-Conditioned Knowledge Distillation" can be an important subsection of SCX-Theory (Paper 2) or a standalone workshop paper.